EBN 111 Class Test 1 (15 March 2024)

Memo

Question 1 to 3

Q1) $Q_b = \int_0^{t_0} i(t)\,dt$ = Area under $i(t)$ from 0h to 8h

$= 1 \cdot 8 \cdot 60 \cdot 60$

$= 28800$ C

Q2) $P(t) = v(t) \cdot i(t) = v(t) \cdot 1 = v(t)$

$W_b = \int_0^{t_0} P(t)\,dt = \int_0^{t_0} v(t)\,dt$ = Area under $v(t)$ from 0h to 8h

$= 10 \cdot 8 + \frac{1}{2} \cdot 80 \cdot 8$

$= 80 + 320$

$= 400$ Wh

Q3) Cost $= W_b \times$ Price

$= 0{,}4$ kWh $\times$ R 2,50 per kWh

$=$ R1,00

Question 4 to 5

KVL : $-12 + 10 I_o - 0{,}5 V_o + 5 - V_o = 0$

$-7 + 10 I_o - 1{,}5 V_o = 0$ — ①

Ohm @ 5Ω : $V_o = -5 I_o$ — ②

② in ① : $-7 + 10 I_o + 7{,}5 I_o = 0$

$17{,}5 I_o = 7$

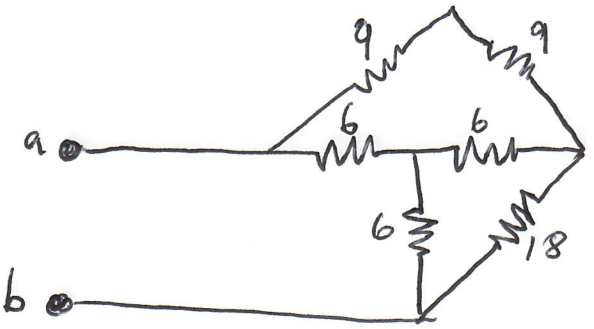
04) $I_o = \frac{7}{17{,}5} = 0{,}4 \text{ A}$

05) $V_o = -5 I_o$

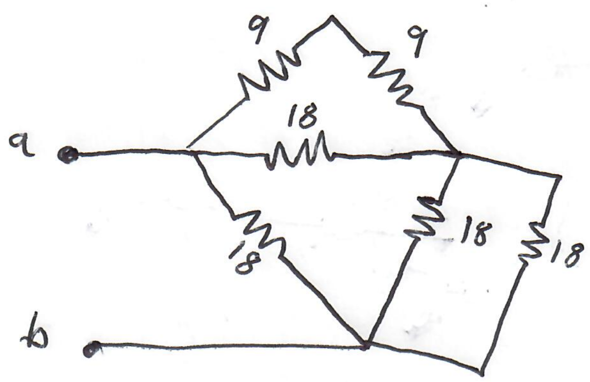
$= -5 \cdot 0{,}4$

$= -2 \text{V}$

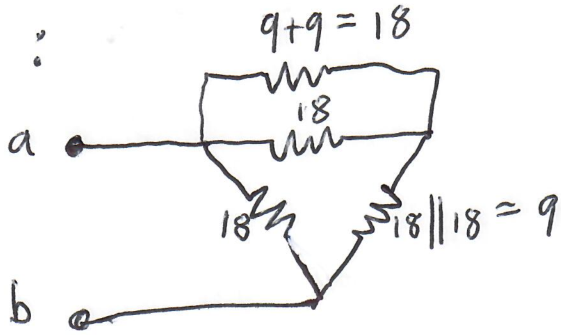
Question 6



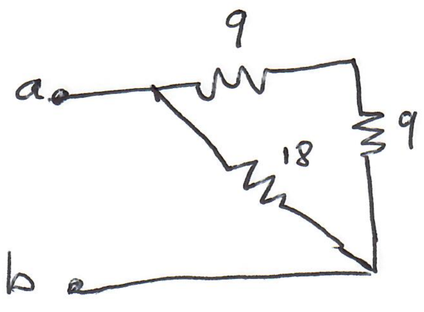
Wye to delta :



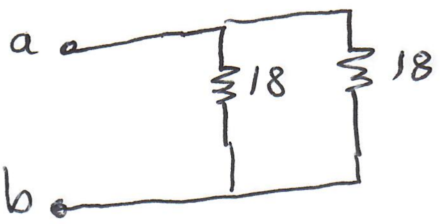
18||18 and 9+9 :



18||18 :



9+9 :



06) $R_{ab} = 9$

Question 7

Nett current flowing into top node:

$$I_1 = 14 - 7$$
$$= 7A$$

Current through 2Ω (use current divider)

$$I_o = I_1 \cdot \frac{5}{(2+5)}$$
$$= 7 \cdot \frac{5}{7}$$
$$= 5A$$

Questions 8 to 11

KCL @ node 1 : $\frac{V_1-10}{10} + \frac{V_1-0}{20} + \frac{V_1-V_2}{5} + 3 = 0$ — ①

① x 20 :

$2V_1 - 20 + V_1 + 4V_1 - 4V_2 + 60 = 0$

$7V_1 - 4V_2 = -40$ — ②

Ohm @ 10Ω : $10 - V_1 = V_0$ — ③

KCL @ node 2 : $V_2 = 5V_0$ — ④

③ in ④ :

$V_2 = 50 - 5V_1$

$\therefore \; 5V_1 + V_2 = 50$ — ⑤

Thus :

$7V_1 - 4V_2 = -40$

$5V_1 + V_2 = 50$

08) $a = 7$

09) $b = -40$

10) $c = 5$

11) $d = 1$